

# Saturn prototype e-receipt solution.

The most important part is not really the e-receipts (they were just smacked together), but the e-receipt delivery method as outlined in: <https://cyberphone.github.io/doc/defensive-publications/e-receipts.pdf>

The receiptUrl is just an example but is a pretty common method to provide secure private access to a specific document without requiring additional authentication.

<https://test.webpki.org/saturn/receipts/2025020700000001/IMo28h6rbCXnnHNoNyaew>

Web-oriented layout where elements like "Core Receipt Data" are individually scrollable *horizontally*. Adaptable elements are possible but not very pretty 😊

The screenshot displays a web browser window with the URL <https://test.webpki.org/saturn/receipts/2025020700000001/IMo28h6rbCXnnHNoNyaew>. The page header shows browser tabs for Java 21, JavaEE 8, and LinkedIn activity. The main content area features the 'Space Shop' logo and contact information: 2000 Avenue des Champs-Élysées, 75000 Paris, France; Phone: 01.50.10.42.08; e-mail: support@spaceshop.com. Below this is the 'Core Receipt Data' table:

| Payee Name | Total      | Subtotal   | Tax              | Time Stamp              | Reference Id     |
|------------|------------|------------|------------------|-------------------------|------------------|
| Space Shop | 636.75 EUR | 530.49 EUR | 106.09 EUR (20%) | 2025-02-07 14:30:58 UTC | 2025020700000001 |

Below the Core Receipt Data is the 'Order Data' table:

| Description                                        | Quantity | Subtotal   |
|----------------------------------------------------|----------|------------|
| Model Rocket<br>SpaceX Falcon Heavy<br>Scale 1:200 | 1        | 499.99 EUR |
| Nasa T-Shirt<br>Grey, Size: L                      | 2        | 30.50 EUR  |

Below the Order Data is the 'Shipping' table:

| Description   | Cost     |
|---------------|----------|
| Free shipping | 0.00 EUR |

Below the Shipping table is a barcode with the reference ID 2025020700000001.

Below the barcode is the 'Return Policy' section:

Return Policy:  
Items can be returned within 30 days of receipt of delivery using the Online Returns Center. Once the item is received at our Customer Support Center, it takes 2 business days for the refund to be processed and 3-5 business days for the refund amount to show up in your account.

Below the Return Policy is the 'Payment Details' table:

| Provider Name | Account Type | Account Id | Transaction Id | Time Stamp              | Request Id       |
|---------------|--------------|------------|----------------|-------------------------|------------------|
| My Bank       | BankNet2     | FR*0404    | #0100345668    | 2025-02-07 14:30:58 UTC | 2025020700000001 |

At the bottom left of the page is a link: [Switch to JSON view](#).

Note that Saturn was created long before the ARF and the EUDIW project! I'm currently in the process of converting from JSON to Deterministically Encoded CBOR.

<https://cyberphone.github.io/wallet-core/doc/>

## Receipt in JSON format

```
{
  "@context": "https://webpki.github.io/saturn/v3",
  "@qualifier": "Receipt",
  "status": "AVAILABLE",
  "referenceId": "2025020700000001",
  "timestamp": "2025-02-07T14:30:58+00:00",
  "commonName": "Space Shop",
  "physicalAddress": ["2000 Avenue des Champs-Élysées", "75000 Paris", "France"],
  "phoneNumber": "01.50.10.42.08",
  "emailAddress": "support@spaceshop.com",
  "amount": "636.75",
  "currency": "EUR",
  "subtotal": "530.49",
  "tax": {
    "amount": "106.09",
    "percentage": "20"
  },
  "shipping": {
    "description": ["Free shipping"],
    "amount": "0.00"
  },
  "barcode": {
    "type": "CODE_128",
    "value": "2025020700000001"
  },
  "freeText": [
    "Return Policy:",
    "Items can be returned within 30 days of receipt of delivery using the Online Returns Center. Once the item is received at our Customer Support Center, it takes 2 business days for the refund to be processed and 3-5 business days for the refund amount to show up in your account."
  ],
  "lineItems": [
    {
      "description": ["Model Rocket", "SpaceX Falcon Heavy", "Scale 1:200"],
      "quantity": "1",
      "subtotal": "499.99"
    },
    {
      "description": ["Nasa T-Shirt", "Grey, Size: L"],
      "quantity": "2",
      "subtotal": "30.50"
    }
  ],
  "paymentMethodName": "BankNet2",
  "accountReference": "FR*0404",
  "payeeAuthorityUrl": "https://test.webpki.org/saturn-payeebank/payees/86344",
  "payerProviderData": {
    "commonName": "My Bank",
    "providerAuthorityUrl": "https://test.webpki.org/saturn-payerbank/authority",
    "referenceId": "#0100345668",
    "payeeRequestId": "2025020700000001",
    "timestamp": "2025-02-07T14:30:58+00:00"
  },
  "receiptSignature": {
    "algorithm": "ES256",
    "publicKey": {
      "kty": "EC",
      "crv": "P-256",
      "x": "8VY09NWUy-aVGNHZZQDIyy-H3RxLFxbiPR2SVIEubjE",
      "y": "OuHehTNjMbphW0s3nBBVdAALLdzE9x-hup4CnJ1gM-o"
    },
    "value": "AbRG__PPShP4R0UTje-JO-8Epju60B9PxV4bfS3_e_Syp3ce-bj-bxIw0gEXFhHG1umYjiPnrXyIhB2A4tuSg"
  }
}
```

[Switch to normal view](#)

Here we have a core problem: how do you verify the authenticity of the signature key? A universal method could be to publish the (public) key on the merchant Web. Decentralization rulez!